3.1.1 What is the body made of?

Students should have knowledge and understanding of the following content.

Content	Additional guidance and suggested TDAs	Specification reference GCSE Combined Science: Trilogy	Specification reference GCSE Combined Science: Synergy
Outcome 1			
Cells are the basic building blocks of all living organisms.		4.2.1	4.1.3.2

Content	Additional guidance and suggested TDAs	Specification reference GCSE Combined Science: Trilogy	Specification reference GCSE Combined Science: Synergy
Most human cells are like most other animal cells and have the following parts: **1.** nucleus – controls the activities of the cells and contains the genetic material **2.** cytoplasm – where most chemical reactions take place **3.** cell membrane – controls the passage of substances in and out of cells. Cells may be specialised to carry out a particular function, eg sperm cells, nerve cells and muscle cells. Students should be able, when provided with appropriate information, to explain how the structure of different types of cell relates to their function.	Details of sub-cellular structures are not needed.	4.1.1.3	4.1.3.2
Outcome 2			
A tissue is a group of cells with a similar structure and function. Students should develop some understanding of size and scale in relation to cells, tissues, organs and systems.		4.2.1	4.2.1.2

Content	Additional guidance and suggested TDAs	Specification reference GCSE Combined Science: Trilogy	Specification reference GCSE Combined Science: Synergy
Organs are aggregations of tissues performing similar functions. Organs are organised into organ systems which work together. Students should be able to identify the position of the major organs in the human body such as the brain, heart, liver, lungs, kidneys and reproductive organs. Students should be able to identify the function of the main organ systems.	Diagram in support booklet (human hair to cell).	4.2.1	4.2.1.2
The human circulatory system consists of the heart, which pumps blood around the body (in a dual circulatory system) and blood, which transports oxygen, proteins and other chemical substances around the body. Students should be able to recognise the different types of blood cell from a photograph or diagram.		4.2.2.2	4.2.1.3
Outcome 3			

Content	Additional guidance and suggested TDAs	Specification reference GCSE Combined Science: Trilogy	Specification reference GCSE Combined Science: Synergy
The human digestive system contains a variety of organs: • salivary glands • stomach • liver • gall bladder • pancreas • small intestine • large intestine. Students should be able to identify the position of these organs on a diagram of the digestive system. Enzymes are used to convert food into soluble substances that can be absorbed into the bloodstream.	Details of specific enzyme reactions are not required.	4.2.2.1	4.2.1.5